Pergamon

Computers Math. Applic. Vol. 34, No. 1, pp. 1–9, 1997
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0898-1221/97 $17.00 + 0.00

PII: S0898-1221(97)00093-X

A Two-Step Jacobi-Type Iterative Method

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(Received August 1995; revised and accepted March 1997)

Abstract—In multigrid methods, it is preferred to employ smoothing techniques which are convergent. In practice, the standard Jacobi and Gauss-Seidel methods are the choices for "smoothers". However, it is known that if the spectral radius condition is violated, then convergence is not guaranteed for these methods. In this paper we develop a simple two-step Jacobi-type method which has better convergence properties and which can be employed as a convergent "smoother" wherever the standard iterative methods fail. We provide the convergence proofs and demonstrate the applicability of the method on a variety of problems.

Keywords—Two-step Jacobi-type method, Multigrid smoother, Convergent conditions proofs, Extended convergence region.

1. INTRODUCTION

In recent years, there has been much interest in multigrid methods because of their accelerated convergence properties [1,2]. Multigrid algorithms are iterative solvers and are applied, in general, to large systems of linear equations obtained from the discretization of partial differential equations on either a finite difference or a finite element mesh. A multigrid algorithm is formulated in such a way that it handles both the high and low frequency errors in the approximate solution. This involves the use of a "smoother" to eliminate high frequency errors and employing the same "smoother" again on coarser grids to eliminate low frequency errors.

In practical applications, a simple iterative method such as the Jacobi or the Gauss-Seidel method is normally chosen as the "smoother" [3]. Although, the multigrid algorithm may work even with a divergent "smoother", provided it diverges slowly and smoothes the error rapidly, practitioners always prefer employing smoothing methods which are convergent. This means that if either the Jacobi or the Gauss-Seidel method is the convergent "smoother", then the coefficient matrix of the system of linear equations should be such that the respective iteration matrix has a spectral radius less than unity [4,5]. So, if the spectral radius condition is violated in either case, one will not prefer to employ the Jacobi or the Gauss-Seidel method as a "smoother". Therefore, it is important to develop simple iterative algorithms which can be applied to systems of linear equations, where the standard iterative methods fail, and are easily employable as convergent "smoothers" in multigrid algorithms. Further, the construction of such iterative algorithms can also be very useful in the study of convection dominated diffusion problems.

Recently, some work along these lines has appeared in the literature [6,7]. In [6], a Gauss-Seidel-type spectrum enveloping technique was developed and in [7], an algorithm based on inner/outer iterations was proposed. The key idea behind the algorithm in [7] was to make the coefficient matrix of the linear system of equations diagonally dominant so that the Jacobi iterative process

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can converge. The two-step Jacobi-type iterative method we construct in this chapter is somewhat similar to the algorithm in [7]. However, by introducing a judicious choice of transformation in the coefficient matrix we obtain a better convergence region compared to that of [7]. We present the convergence proofs and demonstrate the usefulness of our method on several test problems where either the standard iterative method or the method in [7] fails.

2. THE TWO-STEP ITERATIVE METHOD

Given the system of equations

$$Ax = b, \tag{2.1}$$

where $A = [a_{ij}]_{i,j=1,\dots,n}$, $b = (b_1, \dots, b_n)^\top$, $x = (x_1, \dots, x_n)^\top$, let us decompose the nonsingular matrix A as $L + D + U$, where L is a strictly lower triangular matrix, D is a diagonal matrix such that $a_{ii} \neq 0$, $i = 1, \dots, n$, (i.e., D is nonsingular) and U is a strictly upper triangular matrix. Now, we can multiply A by D^{-1} in order to get every element in the main diagonal to be equal to 1. *Therefore, without loss of generality, we consider coefficient matrices A where D is simply I, the identity matrix.*

Let $r = \rho(L + U)$, the spectral radius of $L + U$. Assuming $D = I$, we rewrite equation (2.1) as

$$[(\alpha + r)I + L + U]x = b + (\alpha + r - 1)x, \tag{2.2}$$

by adding the vector $(\alpha + r - 1)x$ to both sides of the equation, where α is a positive parameter.

Let $x^{(0)}$ (the initial guess) be given. Define the sequence $\{x^{(m)}\}_{m=0,1,\dots}$ by

$$[(\alpha + r)I + L + U]x^{(m+1)} = b + (\alpha + r - 1)x^{(m)}. \tag{2.3}$$

Then the sequence $\{x^{(m)}\}_{m=0,1,\dots}$ converges if the spectral radius $\rho(M(\alpha)) < 1$, where

$$M(\alpha) = (\alpha + r - 1)[(\alpha + r)I + L + U]^{-1}. \tag{2.4}$$

But now, since the unknown x is also on the right-hand side of equation (2.2), for each m, we need to solve a system of the form

$$\tilde{A}x^{(m+1)} = \tilde{b},$$

where

$$\tilde{A} = [(\alpha + r)I + L + U]$$

and

$$\tilde{b} = b + (\alpha + r - 1)x^{(m)}.$$

This can be done by using the Jacobi iterative method.

Hence, solving the system of equations $Ax = b$ involves employing the Jacobi iterative method on systems of the form $\tilde{A}x^{(m+1)} = \tilde{b}$, $\forall\, m$ and then generating the sequence $\{x^{(m)}\}$ corresponding to (2.3).

The algorithm of the two-step Jacobi-type iterative method we propose can be given as follows.

ALGORITHM 2.1.

Given ϵ_1, ϵ_2 (> 0) the tolerances for the outer and the inner iterations,
respectively, and an initial vector $x^{(0)}$
Let $error_1 = 1$
Let $m = 0$
Do while ($error_1 \geq \epsilon_1$)
Let $error_2 = 1$
Let $k = 0$
Let $\tilde{b} = b + (\alpha + r - 1)x^{(m)}$

Let $x_{(k)} = x^{(m)}$
Do while ($error_2 \geq \epsilon_2$)

$$x_{(k+1)} = \frac{1}{\alpha + r}\tilde{b} - \frac{1}{\alpha + r}(L + U)x_{(k)}$$

$$error_2 = \|x_{(k+1)} - x_{(k)}\|$$

$$k = k + 1$$

End do
$x^{(m+1)} = x_{(k+1)}$
$error_1 = \|x^{(m+1)} - x^{(m)}\|$
$m = m + 1$
End do

3. CONVERGENCE OF THE ITERATIVE METHOD

It can be seen that if $r < 1$ and α is chosen such that $\alpha + r = 1$, then the proposed algorithm reduces to the Jacobi iterative method, and hence converges. Therefore, it is sufficient to study the case when $r \geq 1$.

Following [7], we will call the iteration defined by equation (2.3) the outer iteration and the iteration given by the Jacobi method the inner iteration. Then we have the following theorems.

THEOREM 3.1. *The inner iteration converges for $\alpha > 0$.*

PROOF. Consider the system

$$\tilde{A}x = \tilde{b}.$$

Applying the Jacobi method, we have

$$x_{(k+1)} = \frac{1}{\alpha + r}\tilde{b} - \frac{1}{\alpha + r}(L + U)x_{(k)}, \qquad k = 0, 1, 2 \ldots.$$

The iteration matrix is given by

$$J(\alpha) = -\frac{1}{\alpha + r}(L + U)$$

with the spectral radius

$$\rho(J(\alpha)) = \frac{1}{\alpha + r}\rho(L + U) = \frac{r}{\alpha + r} < 1. \tag{3.1}$$

Therefore the inner iteration converges for any $\alpha > 0$. The larger α is, the faster the convergence will be.

THEOREM 3.2. *Let $\mu_j = \beta_j \pm i\,\gamma_j$, $(i = \sqrt{-1})$, $\gamma_j \geq 0$, $j = 1, \ldots, n$ be the eigenvalues of $L + U$. Then, the spectral radius*

$$\rho(M(\alpha)) = \frac{\alpha + r - 1}{\min\limits_{j=1,\ldots,n} \sqrt{(\alpha + r + \beta_j)^2 + \gamma_j^2}}. \tag{3.2}$$

PROOF. Let $\mu_j = \beta_j \pm i\,\gamma_j$, $j = 1, \ldots, n$ be the eigenvalues of $L + U$ and η be an eigenvalue of $M(\alpha)$ with a corresponding eigenvector x.

Then,

$$(\alpha + r - 1)[(\alpha + r)I + L + U]^{-1}x = \eta x$$

or

$$(L + U)x = \left[\frac{\alpha + r - 1}{\eta} - (\alpha + r)\right] x.$$

So, we can relate $[((\alpha + r - 1)/\eta) - (\alpha + r)]$ to a μ_j for some j. Then

$$\eta = \frac{\alpha + r - 1}{\alpha + r + \mu_j}$$

and

$$\rho(M(\alpha)) = \max_{j=1,\dots,n} \left| \frac{\alpha + r - 1}{\alpha + r + \mu_j} \right|$$

or

$$\rho(M(\alpha)) = \frac{\alpha + r - 1}{\min\limits_{j=1,\dots,n} \sqrt{(\alpha + r + \beta_j)^2 + \gamma_j^2}}.$$

Given that $L + U = A - I$, if μ is an eigenvalue of $L + U$, then $\lambda = 1 + \mu$ is an eigenvalue of A. Moreover, since A is nonsingular, we cannot have $\operatorname{Re}\mu = -1$ and $\operatorname{Im}\mu = 0$ simultaneously.

Based on the above facts, we can write down the following corollaries. Wherever the proofs are too simple or obvious, we do not present them.

COROLLARY 3.1. *Let $\mu_j = \beta_j \pm i\,\gamma_j$, $\gamma_j \geq 0$, $j = 1, \dots, n$ be the eigenvalues of $L + U$. Then the outer iteration converges if $-1 \leq \beta_j$, $j = 1 \dots, n$.*

If $r = 1$, then we have $-1 \leq \beta_j$, $j = 1, \dots, n$, then it follows from Corollary 3.1 and Theorem 3.1 that Algorithm 2.1 converges for $r = 1$. Therefore, we will assume that $r > 1$.

COROLLARY 3.2. *Let $\lambda_j = \delta_j \pm i$, $\kappa_j, \kappa_j \geq 0$, $j = 1, \dots, n$ be the eigenvalues of A. Then the outer iteration converges if $0 \leq \delta_j$, $j = 1 \dots, n$.*

COROLLARY 3.3. *The outer iteration converges if A is positive definite.*

From Corollary 3.2 and the Gerschgorin Theorem [8,9], we have the following corollary.

COROLLARY 3.4. *If A is such that $\sum_{\substack{j=1 \\ j \neq i}}^{n} |a_{ij}| = 1$, $i = 1, \dots, n$, then the outer iteration converges.*

PROOF. The Gerschgorin Theorem states that all the eigenvalues of A are in the set $S = \cup_{l=1}^{n} S_l$, where

$$S_l = \left\{ z \in C; |z - 1| \leq \sum_{\substack{j=1 \\ j \neq l}}^{n} |a_{lj}| \right\}.$$

Given that $\sum_{\substack{j=1 \\ j \neq i}}^{n} |a_{ij}| = 1$, then all the eigenvalues of A lie inside the circle $|z - 1| = 1$, where $z = (x + i\,y) \in C$, and hence, they have nonnegative real parts.

If A is skew-Hermitian, then all the eigenvalues are purely imaginary and we have another corollary.

COROLLARY 3.5. *The outer iteration converges if A is skew-Hermitian.*

In the case of the tridiagonal matrix, $\operatorname{tridiag}(a, 1, b)$, we can show the result given below in Corollary 3.6.

COROLLARY 3.6. *Let*

$$A = \begin{bmatrix} 1 & b & & & & \\ a & 1 & b & & & \\ & a & 1 & b & & \\ & & \ddots & \ddots & \ddots & \\ & & & a & 1 & b \\ & & & & a & 1 \end{bmatrix} \in \Re^{n \times n},$$

then the outer iteration converges if $ab \leq 1/4$.

PROOF. The eigenvalues of A are given by

$$\lambda_k = 1 + 2(ab)^{1/2} \cos\left(\frac{k\pi}{n+1}\right), \qquad k = 1, \ldots, n.$$

If $ab > 0$, then λ_k is real for every k. Since $\cos(\frac{k\pi}{n+1}) > -1$, $k = 1, \ldots, n$, we have that $\lambda_k > 1 - 2(ab)^{1/2}$. Therefore, if $1 - 2(ab)^{1/2} \geq 0$, then $\lambda_k > 0$, and consequently, the outer iteration converges if $ab \leq 1/4$.

If $ab < 0$, then $\operatorname{Re} \lambda_k = 1 (> 0)$ for all k, and therefore, the outer iteration will converge.

The conditions for convergence obtained so far depend only on the real parts of the eigenvalues of $L+U$. However, it is possible to obtain conditions for convergence which involve the imaginary parts of the eigenvalues (of $L+U$), too. The following theorem gives such a condition.

THEOREM 3.3. *If all the eigenvalues of $L+U$ are outside the circle, $|z+r| = r-1$, then there exists a $\alpha_0 > 0$ such that Algorithm 2.1 converges for all $\alpha \in (0, \alpha_0)$.*

PROOF. From (3.2), we have that $\rho(M(\alpha)) < 1$ if

$$\frac{(\alpha + r - 1)^2}{(\alpha + r + \beta_j)^2 + \gamma_j^2} < 1.$$

If $\beta_j \geq -1$, then the eigenvalues are outside of the circle $|z+r| = r-1$ and we know that the above inequality holds. So, we study only those eigenvalues for which $\beta_j < -1$. Consider

$$\begin{aligned}(\alpha + r + \beta_j)^2 + \gamma_j^2 &= (\alpha + r - 1 + \beta_j + 1)^2 + \gamma_j^2 \\ &= (\alpha + r - 1)^2 + 2(\alpha + r - 1)(\beta_j + 1) + (\beta_j + 1)^2 + \gamma_j^2.\end{aligned}$$

We will have $\rho(M(\alpha)) < 1$ if

$$2(\alpha + r - 1)(\beta_j + 1) + (\beta_j + 1)^2 + \gamma_j^2 > 0$$

or

$$\alpha < \frac{-2(r-1)(\beta_j + 1) - (\beta_j + 1)^2 - \gamma_j^2}{2(\beta_j + 1)} = \alpha_0, \text{ say.}$$

In order to choose an α (> 0) according to the above inequality, we need to have $\alpha_0 > 0$, i.e., we should have the condition

$$(\beta_j + 1)^2 + 2(r-1)(\beta_j + 1) + \gamma_j^2 > 0,$$

or equivalently,

$$(\beta_j + r)^2 + \gamma_j^2 > (r-1)^2.$$

This means that the algorithm will converge if the eigenvalues of $L+U$ are outside the circle $|z+r| = r-1$ and α is chosen such that $0 < \alpha < \alpha_0$.

In the following section, we present some results obtained by carrying out numerical experiments using the proposed two-step method. It will be seen that the iterative method developed here has an extended convergence region than the algorithm in [7].

4. NUMERICAL RESULTS

We performed the numerical experiments on four different linear systems of equations $A_i x = b_i$ $(i = 1, \ldots, 4)$. The initial vectors for the outer and the inner iterations were $x^{(0)} = (0\ 0\ \ldots\ 0)^\top$ and $x_{(0)} = x^{(m)}$, respectively. The respective stopping criteria of the iterations were $\|x^{(m+1)} -$

$x^{(m)}\|_2 < \epsilon_1$ and $\|x_{(k+1)} - x_{(k)}\|_2 < \epsilon_2$. In all the computations, a Fortran implementation of the method was used.

The matrices A_i of the systems $A_i x = b_i$ are given by

$$A_1 = \begin{bmatrix} 1 & 1 & 0 \\ -2 & 1 & 1 \\ 0 & -2 & 1 \end{bmatrix},$$

$$A_2 = \begin{bmatrix} 1 & \frac{5}{23} & \frac{7}{23} & \frac{14}{23} & \frac{16}{23} \\ \frac{17}{24} & 1 & \frac{1}{24} & \frac{1}{3} & \frac{5}{8} \\ \frac{11}{25} & \frac{18}{25} & 1 & \frac{2}{25} & \frac{9}{25} \\ \frac{10}{21} & \frac{4}{7} & \frac{19}{21} & 1 & \frac{1}{7} \\ \frac{2}{11} & \frac{3}{11} & \frac{13}{22} & \frac{10}{11} & 1 \end{bmatrix},$$

$$A_3 = \begin{bmatrix} 1 & 5 & 4 & 4 & 3 \\ \frac{1}{4} & 1 & -\frac{1}{2} & 1 & -\frac{1}{4} \\ 0 & \frac{2}{3} & 1 & \frac{2}{3} & \frac{1}{3} \\ \frac{3}{10} & \frac{8}{10} & \frac{3}{10} & 1 & \frac{9}{10} \\ 1 & 4 & 1 & 2 & 1 \end{bmatrix},$$

and

$$A_4 = \begin{bmatrix} 1 & 1 & 1 & 3 \\ -1 & 1 & -1 & 1 \\ -1 & -1 & 1 & 1 \\ 1 & -1 & 1 & 1 \end{bmatrix}.$$

The corresponding b_i are

$$b_1 = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \qquad b_2 = \begin{bmatrix} \frac{65}{23} \\ \frac{65}{24} \\ \frac{13}{5} \\ \frac{65}{21} \\ \frac{65}{22} \end{bmatrix}, \qquad b_3 = \begin{bmatrix} 17 \\ \frac{3}{2} \\ \frac{8}{3} \\ \frac{33}{10} \\ 9 \end{bmatrix}, \qquad \text{and} \qquad b_4 = \begin{bmatrix} 6 \\ 0 \\ 0 \\ 2 \end{bmatrix}.$$

For the system given by $i = 2$, the Gauss-Seidel method converges, and for the systems corresponding to $i = 1, 2$, and 3, the method in [7] converges. For the system $A_4 x = b_4$, neither Gauss-Seidel nor the algorithm in [7] converges, but the method we propose converges. This demonstrates that our method has a better convergence region.

In Tables 1 and 2, we present the eigenvalues of the matrices $(L + U)$ and $(I + U)^{-1}L$ with respect to the test problems and the associated spectral radii.

Table 1. Eigenvalues of the iteration matrix $L + U$.

	Eigenvalues of $L+U$	Spectral Radius $\rho(L+U)$	Convergence Jacobi
A_1	0 $\pm 2i$	2	no
A_2	1.8383 $-0.4649 \pm 0.7578i$ $-0.4542 \pm 0.1663i$	1.8383	no
A_3	3.5607 $-0.8572 \pm 0.1112i$ $-0.9230 \pm 0.8147i$	3.5607	no
A_4	1 1.5943 $-1.2972 \pm 1.2056i$	1.7709	no

Table 2. Eigenvalues of the iteration matrix $(I + U)^{-1}L$.

	Eigenvalues of $(I+U)^{-1}L$	Spectral Radius $\rho\left((I+U)^{-1}L\right)$	Convergence Gauss-Seidel
A_1	0 0 4	4	no
A_2	0 $-0.0241 \pm 0.0910i$ -0.0132 -0.1611	0.1611	yes
A_3	0 0.7011 -0.9328 $-1.209 \pm 0.9119i$	1.5144	no
A_4	0 0.0874 $-2.5437 \pm 2.2303i$	3.3830	no

The tolerances for the inner and outer iterations for the systems given by $i = 1, 2, 3$ are $\epsilon_1 = \epsilon_2 = 10^{-5}$. The tolerances for the inner and the outer iterations for $A_4 x = b_4$ are $\epsilon_1 = 10^{-5}$ and $\epsilon_2 = 10^{-7}$, respectively. It was found numerically that if an eigenvalue of the corresponding iteration matrix is outside the circle $|z + r| = r - 1$ with real part less than -1, we must take the tolerance for the inner iteration small enough so that the outer iteration converges.

Figures 1–4 show the relative positions of the eigenvalues of the matrix $(L+U)$ with respect to the circles $|z| = r$, $|z + r| = r - 1$, and the line $\operatorname{Re} z = -1$. (The little closed circles indicate these positions.) It should be pointed out that the algorithm in [7] converges only if the eigenvalues of the matrix $(L + U)$ have real parts which are greater than -1, whereas the method developed in this paper converges even if some of the eigenvalues of the matrix $(L + U)$ have real parts which are less than -1, as demonstrated by the example corresponding to Figure 4.

Although our numerical results are consistent with and illustrate the theoretical points we made in Section 3, the question is, how well can we choose α? From (3.2), we know that $\rho(M(\alpha))$ is an increasing function of α with $\lim_{\alpha\to\infty} \rho(M(\alpha)) = 1$, if $\beta_j \geq -1$. Then, the larger α is, the slower the convergence of the outer iteration will be. Moreover, from (3.1), we have that the large α will give faster convergence of the inner iteration. Thus, there is an optimal value of α

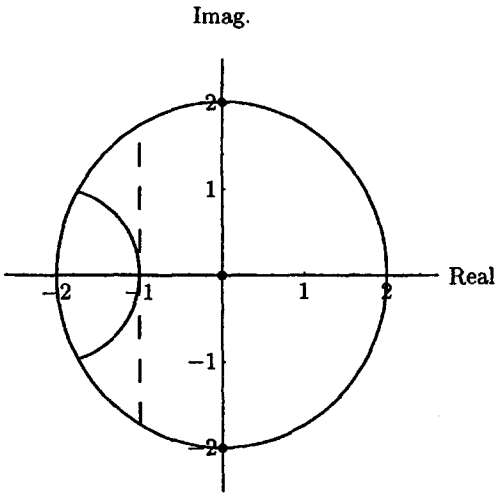

Figure 1. Eigenspectrum of $L_1 + U_1$.

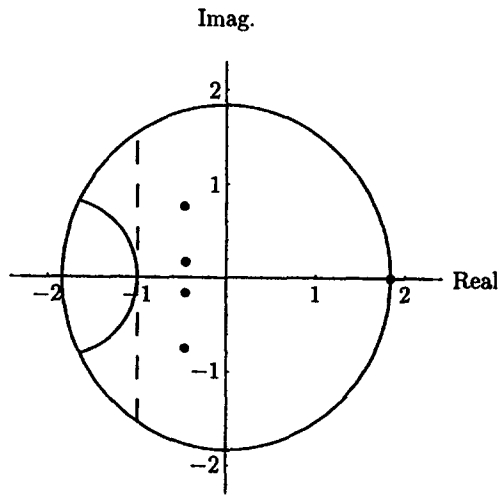

Figure 2. Eigenspectrum of $L_2 + U_2$.

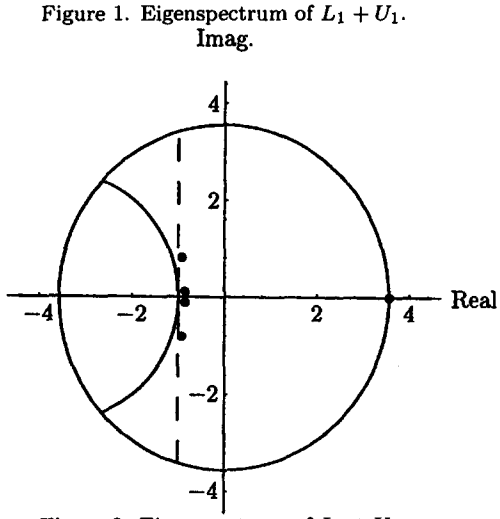

Figure 3. Eigenspectrum of $L_3 + U_3$.

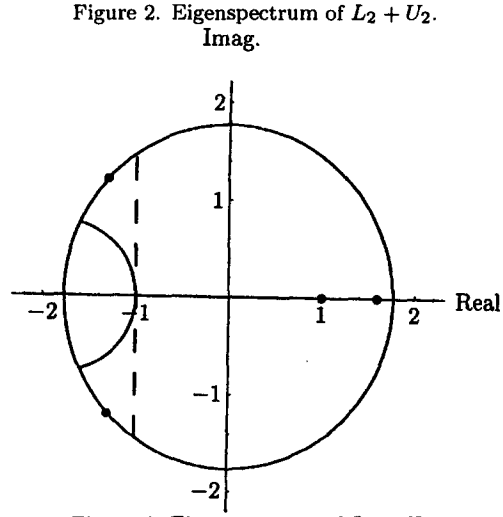

Figure 4. Eigenspectrum of $L_4 + U_4$.

which makes the convergence of our two-step method faster. We determined (numerically) this optimal value of α for the test problems, and in Table 3, we present the number of outer/inner iterations needed for convergence. In Table 4, we give the results when α is chosen such that $\rho(J(\alpha)) = \rho(M(\alpha))$.

Moreover, our extensive numerical study indicates that, choosing a small value for α always leads to convergence at a reasonable rate. Even though there is no theoretical basis, numerical evidence suggests, $\alpha = 1/r$, where r is the estimated value of the spectral radius $\rho(L + U)$, is always a good choice. Further, on problems where the algorithm proposed in [7] works, our two-step method was found to be as competitive as that method.

Table 3. Number of iterations for the optimal α.

Matrix	Optimal α	Outer Iteration	Inner Iteration
A_1	2.2	40	289
A_2	1.9	43	259
A_3	1.1	305	1930
A_4	0.38	59	2552

Table 4. Number of iterations when α is chosen such that $\rho(J(\alpha)) = \rho(M(\alpha))$.

Matrix	α	Outer Iteration	Inner Iteration
A_1	1.0	28	333
A_2	0.6723	26	322
A_3	0.1877	180	3102
A_4	0.4408	54	2565

5. CONCLUDING REMARKS

In this paper, we have developed a simple two-step Jacobi-type iterative method for solving systems of linear equations. We obtained the conditions for convergence for various cases and demonstrated the applicability of the method on problems where the other known simple iterative methods fail. In regard to multigrid techniques, whenever the standard Jacobi and Gauss-Seidel methods fail as convergent "smoothers", this two-step Jacobi-type method can be easily implemented as the new convergent "smoother". Further, this method is also amenable to parallel adaptation.

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